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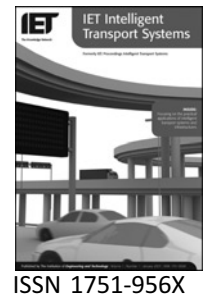
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# A method to assess the driver mental workload: The driving activity load index (DALI)

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**Abstract:** Intelligent transport system (ITS) can induce improvement in road safety, as long as acceptability by user's population and adequacy with driving task demands have been checked. The Human-Centred Design process allows taking into account the diversified driver's population needs and functional abilities in addition to the complexity of the driving context, for setting up design recommendations and for the development of methodologies for safety assessment. Evaluation of the driver's mental workload is an important parameter, complementary to objective ones such as control of the vehicle and driver's visual strategies. This paper describes experiments conducted in the framework of the European project AIDE aiming at validating the DALI (Driving Activity Load Index), a tool set up to allow the evaluation of driver's mental workload while driving, with or without the support of in-vehicle systems. The main results and conclusion from the conducted analysis are presented.

## 1 Introduction

If the driving task has evolved little since the creation of the car, this situation is changing today under the combined effect of widespread driver information and communication systems (IVIS) and the emergence of advanced driver assistance systems (ADAS), covered under the general term of on-board Intelligent Transport System (ITS).

These systems brought a strong hope in terms of improvement in road safety, mobility and transport environment with traffic optimisation, as they allowed an electronic support to human being functional abilities and to road management. Several functions are already available, dealing with drivers' perceptive, cognitive and motor abilities such as preparation to unexpected events, decision taking under time constraint and reaction time in emergency situations. Nevertheless, the driving task is a complex activity, and the system functions have to match with the driver's expectations, needs, requirements and capacities. This really is a challenge when one realises that there is a wide heterogeneity of drivers, meaning that the same product has to fit with an important range of contexts and users. This statement is true for any product, but is

even more challenging in the context of the driving task, owing to its real time constraint and the severity of the issue in terms of road safety. All these considerations lead to conduct investigations about human-centred design process, in order to avoid misconceptions as much as possible, and in order to ensure safety, reliability and acceptability of the proposed functions for a wide range of environments and types of drivers.

Many research studies have been conducted to investigate the road safety consequences of deploying on-board ITS in vehicles [1, 2]. The objective was to evaluate the balance between the potential interference induced by these systems, leading to an increase in critical situations, versus the potential benefits that can be derived from them in supporting the driving task.

To assess the safety of these systems, it is necessary to have an efficient methodology that is applied according to the type of function, the type of system and the context in which the system is used [3]. A quite exhaustive overview of available methodologies, tools and techniques has been conducted within the framework of the network of Excellence HUMANIST [4]. Classically, the parameters that have

been taken into consideration for safety evaluation have been related to the vehicle control; for example, deviations from vehicle trajectories deriving from the system use [5]. Other parameters include drivers' visual strategies, visual demand deriving from the use of on-board screens [6], general driver behaviours [7, 8] and overall driver workload according to the situation [9].

The level of driver's workload is linked to the difficulty of the task he experienced, and to his choice of strategies between the effort and performance trade-off [10]. Indeed, the individual can adapt his behaviour facing an increased task complexity with two possibilities: a higher investment of effort with no perceptible effect on performance; or, on the contrary, no increase in effort leading to a lower level of performance. Then, moderate increases in task difficulty may produce few observable changes in error rate, as the driver attempts to keep performance constant by allocating more resources to the task [11]. Furthermore, inter-individual strategies are variables; some individuals develop more effective strategies which require less effort to reach a level of performance than do others. Hence, for all these reasons, objective performance measures, which directly measure performance, are not sufficient by themselves to evaluate the overall demands of a given situation, data on effort have also to be taken into account.

Mental workload is a psychological construct, difficult to define and difficult to assess. O'Donnell & Eggemeir [12] defined workload as being this portion of the operator's limited capacity that is actually required to perform a particular task. According to this definition, mental workload depends upon the task demands in relation to the amount of resources the operator is willing or able to allocate, and is therefore a relative concept [13]. In order to define workload, the concept of 'effort' is of a primary importance: processing effort in resource allocation and effort for mobilisation of additional resources as a compensatory process, in relation to task demand [14, 15]. Although there is no universally accepted definition of mental workload, a consensus suggests that mental workload can be conceptualised as the interaction between the structure of systems and tasks on the one hand, and the capabilities, motivation and state of the human operator on the other [16–18].

Several methods have been developed to measure mental workload [19]: measurements of physiological parameters, such as heart rate [20], dual-task method [21]; and methods that elicit drivers' subjective judgments about the workload they have experienced. The latter include SWAT (subjective workload assessment technique; [22] and the NASA-TLX – Task Load Index [23]). Subjective measures allow estimates from individual's reports concerning the workload or effort expenditure that was experienced during task [24]. These measures are often used in practice because they have many practical advantages over objective measures [25–27].

The next section of this paper discusses the subjective methods for the evaluation of drivers' mental workload, and, more specifically, on a revised version of the NASA-TLX.

## 2 The DALI as a method for subjective evaluation of mental workload

The subjective method allow for comparative assessment of mental workload in different situations. It can therefore be regarded as a relative measure. Subjective evaluation is often conducted in association with other workload measurement techniques [28].

The SWAT is a sophisticated workload assessment tool, composed of a two-step process: in a scale development phase, data necessary to develop a workload scale are obtained from individuals; during an event scoring phase, individuals rate the workload associated with a particular task [29]. The primary assumption of SWAT is that workload is a function of three dimensions: time load, mental effort load and psychological stress, each dimension having three possible levels. All possible combinations of the three levels of each dimension yield a 27-cell, three-dimensional, matrix to represent workload.

The NASA-TLX method assumes that workload is influenced by mental demand, physical demand, temporal demand, performance, frustration level and effort. After assessing the magnitude of each of these six factors on a scale, the individual performs pairwise comparisons between these six factors, in order to determine the higher source of workload factor for each pair. A composite note quantifying the level of workload is set up by using both factor rating and relative weights computed from the comparison phase.

The NASA-TLX has been tested and used by the army and has been considered superior in terms of sensitivity than other methods and well accepted by the operator [30].

The DALI (Driving Activity Load Index) is a revised version of the NASA-TLX, adapted to the driving task. As previously mentioned, mental workload is multidimensional and, among other things, depends upon the type of loading task. The NASA-TLX was originally designed to assess pilot workload in the aviation domain. The goal was to develop a test better able to assess driving workload.

In this paper, the intention is not to develop principles and modes of scoring of the NASA-TLX, numerous literatures since 1988 have covered this topic. It should be stressed that, except for the definition of the main factors, the basic principles of DALI are the same as those for the TLX. In both tools, a scale rating procedure is defined for six pre-defined factors, followed by a weighting procedure in order to combine the six individual scales into a global

score. The main difference lies in the choice of the main factors composing the workload score, in order to be more adapted to the driving context.

For the NASA-TLX, one of the factors to be rated is called the 'physical demand' component and is usually defined in the following terms: 'How much physical activity was required? – pushing, pulling, turning, controlling, activating, ...'. It appears that this question would not be very relevant when considering the driving activity where the control of the vehicle is quite automatic for an experienced driver, and where maneuvers are not supposed to be physically demanding in modern cars.

Another example is the 'mental demand' component defined in the TLX as follows: 'How much 'mental' and perceptual activity was required? – thinking, deciding, calculating, remembering, looking, searching, ...'. This statement covers both perceptual and cognitive aspects of workload, and it is thought that it would be interesting in the context of the driving task to be able to identify these various modalities (perceptual, cognitive) on the mental workload.

Finally, the evaluation of the 'performance' factor can be made using objective data. The subjective rating of a good performance by the driver can show discrepancies with the measured one, but this difference might be due to many factors other than the mental workload itself – low or high self-esteem, motivations to fit to the standard performance, .... The DALI development began in 1994, the process covered preliminary investigation with interviews with experts, initial items generation, analysis, initial pilot testing, preliminary administration of the test and validation. This investigation led to the development of seven workload dimensions for DALI: effort of attention, visual demand, auditory demand, temporal demand, interference and situational stress (Table 1).

This tool has been used in order to assess driving workload with and without secondary activities, in order to compare the level of load in several conditions. The main results of experiments conducted in real road context, testing the tool validity while using mobile phone and navigation systems, have been published in 1997 [31].

### 3 Analysis of DALI efficiency for driver's workload evaluation in ITS context

An experimentation aimed at validating the DALI method in diversified driving situations varying by their level of complexity was conducted recently [32]. The study has been conducted in the framework of the European project AIDE (Adaptive Integrated Driver-vehicle InterfacE) supported by the DG InfoSo. The general objective of this project was the assumption that if the driver, the vehicle

**Table 1** Factors of the DALI and their description

| Title               | Endpoints | Description                                                                                                                                                               |
|---------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| effort of attention | low/high  | to evaluate the attention required by the activity – to think about, to decide, to choose, to look for and so on                                                          |
| visual demand       | low/high  | to evaluate the visual demand necessary for the activity                                                                                                                  |
| auditory demand     | low/high  | to evaluate the auditory demand necessary for the activity                                                                                                                |
| temporal demand     | low/high  | to evaluate the specific constraint owing to timing demand when running the activity                                                                                      |
| interference        | low/high  | to evaluate the possible disturbance when running the driving activity simultaneously with any other supplementary task such as phoning, using systems or radio and so on |
| situational stress  | low/high  | to evaluate the level of constraints/stress while conducting the activity such as fatigue, insecure feeling, irritation, discouragement and so on                         |

and the environment (DVE state) are monitored, the driver-vehicle interface can be adapted accordingly in order to optimise safety and usability for the driver. Within the objective of designing this adaptative interface, one important part of the process was to define adapted methodology to evaluate developed prototypes through iterative phases. Evaluation of the driver's workload was part of the methodology and tools to be used for this purpose have to be tested first.

A specific experiment has been developed in this context to test the DALI method. The hypothesis of this experiment were:

- The DALI factors will have high values for complex driving conditions, including use of in-vehicle system or not, as these conditions will induce high workload for the driver.
- The values of the DALI factors will be consistent with the characteristics of the driving context (e.g. high values for the DALI auditory factor will be recorded in driving context where complex auditory messages are displayed).

### 3.1 Study design

As the objective of the experiment was to test the method, the driving conditions have been set up to induce on purpose various levels of workload for the driver. Indeed, various complexities of the context will allow to evaluate if the tool reflects correctly what is expected according to the driving conditions, and in which way the data resulting from the subjective evaluation correspond to the level of workload deliberately induced on the driver.

The four experimental sessions, of 15 min each, conducted on real road presented the following characteristics:

- to vary according to the level of workload induced on the driver;
- to be as realistic as possible in a context of the driving task.

**Two situations with a high-task demand:** *High (context + system) HCS*: While driving, the driver had to run a task according to stimulations emitted by an on-board system. The information to deal with is not directly linked to the driving task and induced a manual action or a verbal answer for the driver. The route to follow is given by a guidance system. The workload was linked to perceptual processes, decision making and motor and/or verbal output.

*High (context) HC*: Before the experimentation started, the driver had to consult a paper map to learn about and to memorise the route to follow. Then, he can stop anytime to check again the directions. The workload was linked to the mental representation of the route and to memorise it.

**Two situations with a low-task demand:** *Low (context + system) LCS*: The driver had to follow the route according to visual and auditory information given by a guidance system. The workload was linked to perceptual processes but the decision making, and the mental representation/memorisation was lighter than in the previous sessions.

*Low (context) LC*: During the route, the experimenter gave on time and clear directions to be followed by the driver. The workload was linked only with the management of the driving task, without any added activity.

In order to evaluate the efficiency of the DALI as a tool for the assessment of mobile phone use, an experiment was carried out in a real-road context (see [31] for detail of the experimental protocol and results). The objective was to investigate the evaluation of perceptual and cognitive load for the driver in this phoning condition, knowing that specific factors of the DALI were dedicated to these aspects. Results indicated that the global value of mental load increased significantly when phoning and driving in comparison with the reference situation

corresponding to no system use. The load index was significantly high for 'auditory' and 'interference' factors, in addition to 'stress'. The effort of 'attention', although higher than during a simple driving task, does not increase significantly. Hence, in terms of subjective evaluation of the workload, drivers identified the disturbance induced by phoning, through the perceptive channel of audition, and through the management of the driving task which induced stress. By this example, it is possible to illustrate how this tool based upon subjective evaluation and driver awareness can allow for the understanding of mental workload and cost of the task.

Hence, these four situations were identified, from high to low demand: 'complex system' requiring cognitive and perceptive attentional demand, 'paper map' with no system, 'guidance system' correctly designed and 'human co-pilot' giving instructions to the driver.

### 3.2 Equipment

The same instrumented vehicle has been used for the four sessions. For the complex session HCS, a specific software has been developed by our laboratory in order to display various types of auditory and visual information on a system that has been implemented on the dashboard. For the LCS session, a classical navigation system GPS has been implemented on the dashboard. This type of system displayed clear and simple auditory messages (e.g. 'be prepared to turn left' then 'turn left'), in addition to guidance visual displays (directional arrows or schema of roundabouts) concerning driving manoeuvres during the route. The use of this system did not require any specific training.

### 3.3 Subjects and procedure

A group of ten experienced middle-aged drivers have been driving in urban and peri-urban areas. After a period of 15 min of familiarisation with the vehicle, each of them had to go through the four sessions described above, the sequence order of these sessions being randomly organised. The subjective evaluation tools have been applied at the end of each of these four sessions.

To summarise, the process for the experimental procedure was as follows:

- To set up diversified situations varying on purpose by their level of demand: cognitive process (e.g. to memorise the route) and perceptive-motor process (e.g. to run manual action following auditory, visual or tactile stimulations).
- To apply the tool for each of these sessions in order to gather subjective data.
- To check that the highly demand session corresponds to the higher values for the tools and to identify in which way.

### 3.4 Results

The non-parametric test Wilcoxon has been conducted in order to analyse the significance of the difference between experimental sessions.

**3.4.1 Global workload:** There is a significant difference between the four experimental sessions in terms of subjective assessment of workload by the driver when looking at the DALI results (Wilcoxon,  $Z = 3.007$ ,  $p = 0.003$ ;  $Z = 2.224$ ,  $p = 0.026$ ,  $Z = 2.539$ ,  $p = 0.011$ ;  $Z = 3.923$ ,  $p < 0.001$ ).

These sessions were defined with this goal and hence this result is very positive while checking the validity and the sensitivity of this tool.

An analysis of the detail of the results for each factor allows to better identify and understand what are the components of this global score (Table 2).

#### 3.4.2 Workload linked to cognitive components:

**Attention:** There is a significant difference between the high and the low workload sessions in terms of attentional requirements (Wilcoxon,  $Z = 2.840$ ,  $p = 0.005$ ;  $Z = 3.869$ ,  $p < 0.001$ ). In the High contexts, the attention required to interact with the complex on-board system is higher than the one to find his route according to the memorised information, but the difference is not that significant ( $Z = 1.991$ ,  $p = 0.047$ ). In the low context, there is no significant difference in terms of attention between using a guidance system and following the instructions of a co-pilot.

**Interference:** In terms of interference, there is no significant difference between the high context with or without system (between HCS & HC: Wilcoxon,  $Z = 0.471$ ,  $p = 0.638$ ), indicating that navigating with a paper map would be rated as interfering with the driving task as using a very complex in-vehicle system or 'ergonomic mock-up' displaying several messages and there is no significant difference between the low context with or without system (between LCS & LC:  $Z = 1.896$ ,  $p = 0.058$ ), indicating that using a well-designed in-vehicle guidance system is

equivalent in terms of interference with the driving task to be guided by a human co-pilot. Nevertheless, there is a significant difference when comparing high context and low context, indicating, among other things, that navigating with a paper map is more interfering for the driving task than using a guidance system (between HC & LCS:  $Z = 3.037$ ,  $p = 0.002$ , between HCS & LCS:  $Z = 3.662$ ,  $p < 0.001$ ).

**Stress:** There is a significant difference between most of the different types of driving contexts in terms of stress (Wilcoxon,  $Z = 2.382$ ,  $p = 0.017$ ;  $Z = 2.041$ ,  $p = 0.041$ ,  $Z = 3.880$ ,  $p < 0.001$ ), with a lesser value between the high context + system and the high context (Wilcoxon,  $Z = 1.729$ ,  $p = 0.084$ ). The factor stress is reflecting a global evaluation of the task constraint for the driver, and, in a coherent manner, is very low in the situation where the co-pilot is supporting the driver, a bit higher when a guidance system is fulfilling this part, much higher when the driver has to memorise his route and very high when the driver has to manage a secondary task in addition to the driving task.

#### 3.4.3 Workload linked to perceptive components:

**Visual factor:** Considering the visual demand of each session, there is a significant difference between the session with high workload high (context + system) & high (context) and the one with low workload low (context + system) & low (context) (Wilcoxon,  $Z = 3.218$ ,  $p = 0.001$ ;  $Z = 3.95$ ,  $p < 0.001$ ).

The DALI allows to show that there is no significant difference between the two sessions 'using an on-board system displaying complex stimulations' and 'using a paper map to find the route' (Wilcoxon,  $Z = 1.312$ ,  $p = 0.190$ ;  $Z = 1.231$ ,  $p = 0.218$ ). There are also no significant differences between the session 'to be guided by a guidance system' and 'to be guided by an other person'. Taking into account the fact that in both situations, the driver relied on the auditory information coming from the system or from the co-pilot, it is relevant to find no significant visual workload in these two contexts.

**Table 2** Values of level of significance for the Wilcoxon test for each DALI factors

| Wilcoxon | Effort of attention | Visual demand | Auditory demand | Temporal demand | Interference | Stress | Global score |
|----------|---------------------|---------------|-----------------|-----------------|--------------|--------|--------------|
| LC-LCS   | NS                  | NS            | NS              | NS              | NS           | 0.05   | 0.01         |
| LCS-HC   | 0.005               | 0.001         | 0.000           | 0.05            | 0.002        | 0.025  | 0.025        |
| HCS-HC   | 0.05                | NS            | 0.000           | NS              | NS           | NS     | 0.003        |
| LCS-HCS  | 0.000               | 0.000         | 0.000           | 0.005           | 0.000        | 0.000  | 0.000        |
| LC-HC    | 0.005               | 0.001         | 0.000           | 0.05            | 0.000        | 0.025  | 0.025        |
| LC-HCS   | 0.000               | 0.000         | 0.000           | 0.005           | 0.000        | 0.000  | 0.000        |

*Auditory factor:* Considering the auditory demand of each session, a very low value of workload is displayed in the situation where the driver has to memorise his route with a paper map and to find his way based upon the road directions in comparison with the three other situations (significant difference (Wilcoxon,  $Z = 3.954$ ,  $p < 0.001$ ;  $Z = 3.771$ ,  $p < 0.001$ ;  $Z = 3.804$ ,  $p < 0.001$ ). Indeed, in this case, even if the general workload of the situation appeared to be high, the DALI results show that the auditory demand is not involved in this workload.

Furthermore, there is no significant difference between the situation 'using a guidance system' and following instructions from a co-pilot, indicating that the auditory messages coming from the on-board system did not induce a noticeable workload by the driver (Wilcoxon,  $Z = 1.144$ ,  $p = 0.253$ ).

*Tactile factor:* Implementation of vibrations in the seat of the vehicle was a first approach to define if the driver was able to detect this kind of 'unusual' stimulus with accuracy, and if this stimulus was inducing workload. The tactile stimulations were quite well detected and induced a light workload in comparison with situations where this stimulation was non-existence (Wilcoxon,  $Z = 3.703$ ,  $p < 0.001$ ). Nevertheless, this workload is far less important than the one induced by auditory and by visual stimulations for the same session.

### 3.4.4 Workload linked to temporal components:

Like for the global score, stress and attention, the temporal demand is highly different in relation to the type of session (Wilcoxon,  $Z = 1.118$ ,  $p = 0.264$ ;  $Z = 1.556$ ,  $p = 0.120$  (Wilcoxon,  $Z = 2.116$ ,  $p < 0.034$ ;  $Z = 2.843$ ,  $p = 0.004$ ). Indeed, like the other three factors, this factor is revealing a global estimation of the cost of the task. As driving task is under time constraint, it is then not surprising to have a workload value in terms of timing closely linked to the level of the task complexity.

In Fig. 1, the values for each factor and for the global score are displayed for the four experimental sessions.

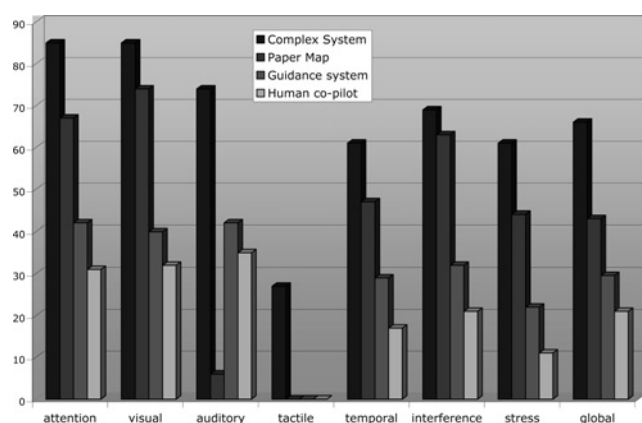


Figure 1 Factors and global value of the DALI

According to the DALI global score, there is a significant difference between the four experimental sessions. More precisely, 'use of guidance instructions' induced generally a lower workload than 'use of a paper map'. Looking at the detail of the DALI factors, it appears that support of the system for the driver is in terms of 'stress', 'interference between driving and finding his route', 'temporal', 'visual' and 'attentional' demand, with significant differences. Of course, 'auditory' demand was not rated by the driver in the context of the paper map use.

Hence, these results demonstrate that a guidance system correctly designed in terms of visual and auditory messages (timing, loudness, legibility, content) is an added value for the driver by making the orientation task lighter in terms of cognitive and perceptive processes than situation with no system. Furthermore, the DALI results showed that there is a higher level of workload while using the system in comparison with relying on the human co-pilot. Hypothesis can be made that this system could require a phase of training longer than the timing of this experiment, in order for the driver to be fully comfortable with the system. Additional testing with a longer training phase could indicate if the system can be equivalent to a human co-pilot or not. At least, the DALI results indicated that using a correctly designed guidance system is superior than using a paper map to find his route.

## 4 Conclusion

The measurement of driver workload complements other workload metrics in bringing additional information and allowing broader understanding about the complex interactions between drivers and the systems they use while driving. DALI, as a subjective workload evaluation tool, allowed for the gathering of data that was usable by the designer in improving his system prototype. It enabled the identification of the impact of a given system implementation by comparing results with a reference situation with no system. One of the main advantages of this tool is that it makes it possible to identify the origins of driver workload, allowing for corrective action at the identified level (e.g., high interference and visual load will indicate that an in-vehicle system has a demanding visual display). The possible design improvement would be to add factors linked to specific aspects of the driving task useful to evaluate the impact of ADAS functions (e.g. level of stress to keep distance with the vehicle ahead, in the case of a system having an impact on this characteristics of the driving task). It is planned to conduct further investigations to improve this method by varying these types of situations. The 'DALI tool kit', comprising the detailed method and procedures for the automatic computation of statistics and the display of graphical outputs, will soon be available on the INRETS web site ([www.inrets.fr](http://www.inrets.fr)), allowing any researcher to use it in his/her scientific context.

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